OXY

State of the Project

Wearable AI assistant · full-stack snapshot

Where fashion meets intelligence

Generated 10 June 2026 · repo: commonsfounder/Oxy · branch: main

SNAPSHOT

Executive Summary

> Oxy has evolved from a product concept with a backend into a real multi-surface assistant stack: an authenticated Cloud Run backend, a PWA, a SwiftUI iOS app, a Gemini voice/chat/image layer, memory, proactive briefings, native phone context, connector actions with safety reviews, and — newly in the repo — pendant firmware for the BLE bridge. Billing, production privacy hardening, and full hardware manufacturing remain the major unfinished pillars.

At a glance

Product	Wearable AI assistant — press-to-talk pendant + phone + cloud; positioned as a “life operating system,” not a chatbot.
Repo	commonsfounder/Oxy (branch **main**) · ~9,500 lines of backend/connector JS, plus SwiftUI app and Arduino firmware.
Backend	Node.js / Express 5 on Google Cloud Run, auto-deployed from GitHub; Supabase (Postgres 17) for data + per-user auth.
Intelligence	Google Gemini — reasoning (gemini-3-flash-preview), live voice (gemini-live-2.5), STT, TTS, image generation.
Surfaces	Backend API · React PWA · native SwiftUI iOS app · BLE pendant firmware (Seeed XIAO nRF52840).
Maturity	Software stack working end-to-end; hardware, billing, and privacy-for-scale still in progress.

WHAT IT IS

Product Definition

Oxy lets a person press a pendant, speak naturally, and have things happen without opening their phone. It uses the phone and cloud as the heavy compute/network layer, and is built around three capabilities working at once.

- **Presence** — be in the moment through press-to-talk voice.
- **Memory** — know the user through stored context and native phone signals.
- **Action** — actually do things through real connected services.

What Oxy is

- A wearable personal assistant
- A hands-free, screen-light interface for daily life
- A connector/action system for real services
- A memory system that learns context over time
- A fashion-tech object with swappable shells

What Oxy is not

- A phone replacement
- Only a note-taker
- A coding assistant
- A generic ChatGPT wrapper
- A camera-first / surveillance device

HOW IT'S BUILT

Architecture & Tech Stack

Layer	Stack	Status
Backend	Node.js, Express 5.1, ws (WebSocket), Multer, Axios, Supabase JS, Sentry	**Live**
AI SDKs	@google/genai, @google/generative-ai (Gemini)	**Live**

Layer	Stack	Status
Data	Supabase / Postgres 17 — per-user auth, RLS migration, indexed tables	**Live**
PWA	Single index.html, React 18 (CDN), service worker, web push	**Live**
iOS	SwiftUI, URLSession + SSE, Keychain, CoreLocation, HealthKit, EventKit, MusicKit	**Live**
Pendant	Arduino / Seeed XIAO nRF52840 Sense — BLE Nordic UART control link	**New in repo**
Deploy	Docker image on Cloud Run; auto-deploy from GitHub; Cloud Run Job for proactive sweeps	**Live**

Required environment

SUPABASE_URL · SUPABASE_KEY · GEMINI_API_KEY · OXY_SESSION_SECRET (plus optional Google OAuth, Telegram, Maps/Places, APNs, Spotify, TransportAPI, Home Assistant).

THE INTELLIGENCE

AI Model Layer

Reasoning / fast	gemini-3-flash-preview (OXY_REASONING_MODEL & OXY_FAST_MODEL)
Live voice	gemini-live-2.5-flash-preview over WebSockets (/realtime-voice, /companion-live)
Speech	Gemini STT + Gemini TTS (preview TTS models, voice playback)
Creative	gemini-2.5-flash-image for image generation; diagrams & presentation artifacts
Grounding	Google Search grounding for live/current facts; never trust training data for changeable info
Efficiency	Prompt caching + context caching; deterministic intent routing before model calls

> Prompt philosophy: a single generalizable-judgment system prompt with per-action guidance, rather than hard-coded routing rules. Tool-argument routing lives next to each action contract.

HIGH-AGENCY AI

Action & Connector System

Every capability is an **action contract** with a risk tier, required fields, and a confirmation mode. High-risk actions (sending messages/email, spending money, placing orders, calls) are **review-gated**: they become a pending-review card and only execute after the user explicitly approves. A connector-health layer turns failures into one-tap recovery (reconnect / retry).

Live connectors

Connector	Capabilities	Mode
Google	Gmail send / read / search; Calendar read / create	**API**
Maps	Find place, directions, trip planning (Google Maps/Places)	**API**
Telegram	Send message, contacts (review-gated send)	**API**
Trainline	Train search, live station board	**API**
Uber	Open ride to a natural destination	**Deep link**
Uber Eats	Open app to a food / restaurant search	**Deep link**

Connector	Capabilities	Mode
Deliveroo	Open app to a food / restaurant search	**Deep link**
Netflix	Title search, add to list	**Deep link**

Internal actions (always available)

find_place · play_music · add_to_music_playlist · generate_visual · create_diagram · create_presentation · forget_memory · reminders / calendar helpers.

> **Recent decision — Uber Eats conversational ordering.** A full server-side ordering flow (9 tools, per-user sessions, DB-backed login, Chromium on Cloud Run) was built against the third-party @striderlabs/mcp-ubereats scraper, then **reverted**: the scraper returns no results against Uber's current site even when authenticated, and there is no official consumer ordering API. Oxy is back to the reliable deep-link handoff. The wiring was sound; the underlying engine was not.

WHERE USERS MEET IT

Product Surfaces

Backend API

Session-authenticated routes for chat & voice (SSE streaming), images/creative artifacts, memory, preferences, connectors (OAuth), actions, native/device context, proactive briefings, history, and debug/health/install.

PWA

Chat, voice recording with realtime fallback, connectors, memory, history, settings; service worker + push notifications.

Native iOS (SwiftUI)

Auth + Keychain session, streaming chat, audio playback, image messages, native local actions (Messages, Calendar, Reminders, Music, Maps), native context (location, health, contacts), proactive tab, connectors, memory, history, settings.

Pendant firmware (new)

Arduino sketch for the Seeed XIAO nRF52840 Sense. Advertises the Nordic UART Service; the app scans by service UUID and reads UTF-8 control commands (START_RECORDING, STOP_RECORDING, TOGGLE_RECORDING, OPEN_CHAT, SEND_MESSAGE, CONFIRM, CANCEL). Voice is captured and transcribed phone-side; the pendant is the press-to-talk control link, not an audio streamer.

DATA & TRUST

Database, Security & Privacy

- **Supabase / Postgres 17** stores users, memories, preferences, connector tokens, action log, and conversations; SQL migrations include auth, indexes, and row-level security.
- **Per-user auth** with session secret; connector tokens stored per user; high-risk actions review-gated; memory is user-visible and editable.
- **New governance docs** in /docs: hardening runbook, privacy-support checklist, onboarding permission matrix.
- **Still needed before scale**: public privacy policy, data export/deletion UX, token encryption beyond DB storage, and a full security review.

REALITY CHECK

Current Working State

Pillar	What exists today	State
Backend	Auth, chat, SSE, Gemini, TTS/STT, image gen, memory, actions, proactive	**Working**
PWA	Chat, voice, connectors, memory, history, settings, push	**Working**
iOS app	Full SwiftUI app with native actions + context	**Working**
Firmware	BLE control-link sketch for nRF52840 pendant	**Early**
Connectors	8 live connectors (API + deep-link)	**Working**
Billing	Production subscription / payment flow	**Not built**
Hardware	CAD, PCB, manufacturing, charger	**Roadmap**
Privacy-for-scale	Legal docs, export/deletion, token encryption	**In progress**

MOMENTUM

Recent Activity

Latest commits on **main**:

- Revert Uber Eats MCP scraper connector — back to reliable deep-link handoff
- Uber Eats conversational-ordering connector built, made Cloud-Run-ready (DB sessions), then reverted
- Close caveats: place ask→reply loop, Uber pickup, Gmail threading
- Directions fixes: personal-place resolution, minute parsing, mode-change follow-ups, arrival/departure derivation
- Timezone fix (run process in Europe/London); incognito (shadow) chat wired end-to-end
- Action cards unified to telemetry style; system prompt rewritten around generalizable judgment

WATCH-OUTS

Main Risks & Roadmap

Key risks

- **Latency** — multi-hop voice round-trips
- **Cost** — per-interaction model + TTS spend
- **Connector reliability** — esp. unofficial / scraped services (see Uber Eats)
- **Hardware feasibility** — bone conduction, battery, BLE bridge
- **Trust** — privacy hardening before real users
- **Platform** — App Store review for an agentic assistant

Roadmap

- **Immediate** — TestFlight, privacy hardening, latency/cost tuning
- **Summer** — Kickstarter narrative + hardware samples, PCBA/CNC quotes
- **H2** — nRF52840 PCB (KiCad/JLCPCB), titanium supplier, charger prototype
- **2027** — fulfil Kickstarter, subscription conversion, shell drops, on-device intelligence

> One-sentence state — A working, multi-surface AI assistant (cloud backend + PWA + iOS + emerging pendant firmware) with real memory, voice, and safe connector actions; the remaining work is hardware production, billing, and privacy hardening for scale.

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